



GSS India

that has the experience of managing diversified products.

How do you manage CO₂ reduction when the product is coming from some other destination by flight to India and then going back by flight to that country?

Let me explain by sharing an actual case study. Bangalore was established as a repair hub for one of the OEMs which collected all these returns from Asia and shipped them to Europe or the US. There was no Asia repair hub back then, but now all the Asian defectives are coming to Bangalore. The distance and time taken to send a product for repair in India is much less as compared to sending it to the US or Europe.

The argument could be why not in-country repairs? However, please keep in mind that not every country can set up their repair shops. There is a huge investment needed to test equipment and invest in capital expenditure. Countries with lower volumes of repairs cannot set up a repair hub. You have to consolidate it and send it to a country where there is a repair hub. In Asia, Bangalore and Futian are strategic repair centers for many of our customers in consumer devices, lifestyle, cloud enterprise and computing (CEC) and industrial segments.

Our customers found that it was not economical to repair some products in Europe and the US because of the labour rates and the time it takes to repair them. India's labour rates are much lower than the US. As such, products that have been declared as beyond economical repair can be repaired here. We are thus reducing CO₂ emissions via our regional repair hub and focussing on reducing scrap by enabling cost-effective repairs.

What are the most common issues in non-circular designs that you encounter and wish the OEM or the design team had thought about this?

The placement of components is very close. Let's take the example of a Ball Grid Array (BGA)—if you keep it too close to some of the other components, it becomes extremely difficult to repair or rework. Sometimes the design is such that they place a certain component which has a higher failure ratio or choose certain components and products from the market which are not reliable.

A lot of components that are being sourced for some products are very customised, and the possibility of circular economy or repairs becomes an issue. In such cases, we recommend to OEMs

to try and make the components more agnostic so that component sourcing does not become a huge challenge. At times, you don't get the specific component that failed and need only that specific one which is not in the market, then the whole product will have to be scrapped. So, we continue working with and advising our customers to think about design for circularity best practices for easier repair and reuse.

Do most of the repair products sent by global customers go back?

The mandatory requirement for any product that comes for repair is that it needs to be returned. It's a customer-related policy, which we strictly follow. So, 100% of these units have to go back including the scrap. If you are getting 100 products with 100 serial numbers, you will need to send these 100 products back, because you cannot scrap or use these products here.

To what factors do you attribute the rising demand for repairing and refurbishing electronic products in companies such as Flex? What other segments will have high growth potential?

The mindset of OEM companies has changed from linearity to circularity, which will bring volumes into companies like Flex, allowing us to repair their products. Then there are environmental and economic advantages of repairing and refurbishing a product, instead of scrapping it. Customers are maximising the value of their products and components in addition to the environmental upside. Reduction in disposals reduces CO₂ and waste. This is going to drive the circular economy and volume, which is our focus.

Emerging products of the future, such as EV batteries, will also create a huge demand in both domestic and international markets. **EFY**